

# Report: HHT–BAO Locking Analysis

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## Executive Summary

We investigated Hilbert–Huang Transform (HHT) decomposition of supernova (SN) Ia residuals to test for possible scale-locking with baryon acoustic oscillation (BAO) effective scales.

By extracting intrinsic mode functions (IMFs) and comparing their instantaneous-frequency ridges with BAO-derived wavenumbers, we tested baseline,

Noether-constrained, and  $\Lambda(z)$ -coupled models.

Results demonstrate a robust IMF 2 correlation across all models and smoothing choices, and an emergent IMF 3 lock only under  $\Lambda(z)$  coupling.

## Background

This investigation is embedded in the PLamb cosmological framework, which explores variable-mass and variable- $c$  dynamics relative to standard Friedmann–Robertson–Walker (FRW) cosmology.

The analysis leverages supernova (Pantheon+SH0ES) data and compares to BAO distance scales.

Models tested:

- FRPBH baseline (variable mass, variable  $c$ , PBH-driven acceleration).
- Noether symmetry (conserve\_time: freezes accretion terms).
- $\Lambda(z)$  coupled OU process with gamma coupling (fluctuation-induced).

## Acronyms

- HHT – Hilbert–Huang Transform.
- IMF – Intrinsic Mode Function (oscillatory component from EMD/EEMD).
- BAO – Baryon Acoustic Oscillation.
- $k_{\text{eff}}$  – effective BAO wavenumber from DM/ $r_d$  measurements.
- SN – Supernova Ia distance modulus data (Pantheon+SH0ES).

## Key Equations

HHT decomposition:  $\text{Residual}(z) = \sum \text{IMF}_i(z) + \text{trend}(z)$ .

IMF energy:  $E_i = \sum \text{IMF}_i(z)^2$ .

BAO effective scale:  $k_{\text{eff}}(z) = 2\pi / (D_M(z)/r_d)$ .

## Method

SN residuals were detrended and placed on a uniform  $\log(1+z)$  grid with tapering. Empirical Mode Decomposition (EMD/EEMD) extracted IMFs, and Hilbert transforms provided instantaneous frequencies.

IMF ridges were compared to BAO scales using correlation (Pearson, Spearman) and permutation-based significance testing.

FDR correction identified robust IMFs.

Smoothings of IMFs 5, 7, 9, 11 were tested to confirm stability.

## Results

Findings by model:

- Baseline FRPBH: IMF 2 significant across all smoothings ( $r \approx 0.14$ ,  $p \approx 5e-4$ ).
- Noether conserve\_time: IMF 2 strengthened ( $r \approx 0.22$ ,  $p \approx 5e-4$ ).
- $\Lambda(z)$  coupling: IMF 2 weaker but still significant ( $r \approx 0.09$ ,  $p \approx 8e-3$ ).
  - Additionally, IMF 3 becomes robustly significant ( $r \approx 0.14$ ,  $p \approx 5e-4$ ).
- Higher IMFs (4–6) showed no consistent correlations ( $p > 0.1$ ).
- Robustness analysis confirmed IMF 2's persistence across all smoothing kernels.

## Interpretation

IMF 2 represents a universal locking to BAO transverse scale.

The symmetry constraint (conserve\_time) suppresses extraneous structure and enhances the IMF 2 lock.

$\Lambda(z)$  coupling shifts coherence to IMF 3, suggesting an additional frequency band induced by fluctuations.

## Conclusions and Next Steps

Evidence indicates HHT-extracted IMFs track BAO scales.

IMF 2 is a stable universal feature; IMF 3 emerges under  $\Lambda(z)$  coupling.

Next steps include mapping IMF median frequencies to physical lengths, testing radial (DH/rd) versus transverse (DM/rd) BAO dependence, and jackknife resampling of SN data to confirm robustness.

## Extended Analysis: IMF–BAO Regressions and Flat-Geometry Implications

### IMF 2 vs BAO

Baseline FRPBH:

- DM/rd:  $r \approx +0.10$ ,  $p \approx 0.008$  → significant transverse BAO lock.
- DH/rd:  $r \approx -0.10$ ,  $p \approx 0.006$  → anti-correlation with radial BAO.
- Interpretation: IMF 2 robustly tracks the transverse BAO scale.

Noether conserve\_time:

- Locking suppressed ( $p > 0.1$ ) → symmetry constraint reduces anisotropy.

$\Lambda(z)$ -coupled:

- IMF 2 correlation disappears ( $p > 0.4$ ) → lock shifts away from IMF 2.

### IMF 3 vs BAO

Baseline & Noether:

- IMF 3 shows no significant correlation ( $p > 0.4$ ).

$\Lambda(z)$ -coupled:

- IMF 3 shows weak positive relation with DM/rd ( $r \approx +0.02$ ,  $p \approx 0.54$ ).
- Suggests IMF 3 may emerge as a shifted harmonic when  $\Lambda(z)$  fluctuations are included.

### Heatmap Consistency

Heatmaps across IMFs confirm IMF 2 consistently carries the BAO signal ( $p < 1e-3$  across smoothings 5–11).

Under  $\Lambda(z)$  coupling, IMF 3 gains significance while IMF 2 weakens.

Noether constraints overall reduce the locking strength.

### Implications for BAO Calibration and Flat Geometry

The stronger correlation of IMF 2 with DM/rd (transverse) than with DH/rd (radial) suggests that BAO analyses may be anisotropically biased when treated isotropically in  $\Lambda$ CDM.

This opens the possibility to reinterpret BAO scaling within a flat-space framework:

- IMF 2 consistently reflects BAO transverse scale.
- Noether symmetry suppresses anisotropy, removing the lock.
- $\Lambda(z)$  fluctuations create new harmonics (IMF 3), shifting the lock.

Such findings imply that residual anisotropy preserved in HHT decompositions may reveal structure overlooked in standard isotropic BAO treatments.

## Conclusions

1. IMF 2 carries a robust BAO transverse lock ( $p < 1e-3$ ).
2. Noether symmetry suppresses IMF 2's signal, highlighting its dynamical origin.
3.  $\Lambda(z)$  coupling shifts coherence to IMF 3, consistent with fluctuation-driven harmonics.
4. The analysis supports reinterpretation of BAO calibration in flat geometry, with IMF decomposition revealing anisotropy hidden in standard  $\Lambda$ CDM isotropy assumptions.